SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.4: Acids, Bases and Salts — Teacher Reference (Pre-filled)

SUMMARY TABLE: GENERAL SCIENCE GRADE 10	
Sub-Strand	Sub-Strand 2.4: Acids, Bases and Salts
Driving Question	DRIVING QUESTION: Why do some substances from our everyday Kenyan environment — lake water, lemon juice, wood ash, cooking salt, antacids — react so differently with metals, food and living things, and how can understanding their acid, base or salt chemistry help us use them safely and sustainably?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Prior Knowledge Elicitation — The Mystery of the Sufuria and the Soda Lake**	Students examined two anchoring phenomena: (1) photographs and short video clips of corroded aluminium sufurias and metal roofing in Kenyan homesteads where lemon juice, tomatoes, or fermented uji had been stored overnight; and (2) satellite images and field photographs of Lake Magadi's vivid pink-white soda flats alongside footage of flamingos feeding there. They also handled labelled sample bottles of lemon juice, milk, wood-ash solution, bicarbonate of soda (baking soda), vinegar, and Lake Magadi water, recording their initial sensory observations (colour, smell where safe, texture of residue). A class Driving Question Board (DQB) was constructed on the wall with students' sticky-note questions grouped into emerging categories: 'What IS in these substances?', 'Why do they react with metals?', 'Are they safe?', and 'How do we measure them?'	Students surfaced and documented their prior conceptions about acids and bases — including common Kenyan folk knowledge (e.g., 'ash water cleans pots because it is strong'; 'lemon juice preserves fish'). They established that substances around them behave very differently and that this difference must have a chemical explanation. The class collectively agreed on the driving question and committed it to the DQB. Pedagogical note: This lesson is purely elicitation — resist correcting misconceptions at this stage. Record every student idea on the DQB without evaluation; this baseline is essential for measuring conceptual change by Lesson 8.	No formal explanation is introduced in Lesson 1. The teacher's role is to provoke curiosity and surface prior knowledge. The phenomena of the corroded sufuria and the soda lake serve as the anchoring context to which every subsequent lesson will return. Teachers should photograph the initial DQB for comparison at the end of the unit. Cold-call at least 6 students using the prompt: 'What do you THINK is causing this — and what makes you say that?' Allow 10-second wait time after each question before accepting responses.
Lesson 2 **pH Investigation:**	Students used universal indicator solution and pH paper (or a digital pH meter where	Substances are classified as acids (pH < 7, dominant H^+ ions), bases (pH > 7,	The anchoring phenomenon is revisited: the sufuria corrodes more in lemon-juice and uji

Measuring the Acid-Base Character of Everyday Kenyan Substances	available) to measure the pH of ten locally sourced substances: lemon juice, uji (fermented porridge), fresh milk, wood-ash solution, bicarbonate of soda solution, Lake Magadi water (simulated with Na_2CO_3 solution), cooking-salt solution, tap water from Nairobi/local source, vinegar, and antacid tablet dissolved in water. Results were recorded in a class data table projected at the front. Students observed that colours ranged from red-orange (acidic) through green (neutral) to blue-purple (alkaline), and noted that Lake Magadi water and wood-ash solution produced the deepest purple, while lemon juice and uji produced the deepest red.	dominant OH^- ions), or neutral (pH = 7, equal H^+ and OH^-). Lake Magadi water is strongly alkaline (pH ≈ 11–12) due to dissolved sodium carbonate (trona/natron). Lemon juice is strongly acidic (pH ≈ 2) due to citric acid. Cooking-salt solution is neutral (pH ≈ 7). Students learned that pH is a logarithmic scale: a change of 1 pH unit represents a tenfold change in H^+ concentration. Pedagogical note: Emphasise that 'acid' and 'base' are not binary labels but positions on a continuous scale — this corrects the common misconception that only laboratory chemicals are acids or bases.	storage (low pH, high H^+) than in plain water storage, because H^+ ions attack the aluminium oxide layer. The pink flamingos of Lake Magadi survive in highly alkaline water because their legs and beaks are adapted — most other organisms cannot. Students update the DQB: the 'What IS in these substances?' cluster now has the answer 'H^+ or OH^- ions in different concentrations.' Teachers should cold-call 4 students to explain why two substances with different colours on pH paper might still both be classified as 'acidic.'
Lesson 3 **Biological and Chemical Roles of Acids and Bases — Digestion, Respiration and Bleaching**	Students watched an annotated diagram walkthrough of the human digestive system, focusing on the stomach (HCl, pH 1–2) and the small intestine (bile + pancreatic juice, pH 7–8). They then examined a demonstration: a piece of nyama choma placed in dilute HCl (simulating gastric acid) versus plain water — observing faster protein breakdown in acid. A second demonstration showed sodium hypochlorite bleach (alkaline) removing colour from a strip of printed fabric, connecting to its use in Kenyan laundries and hospitals. Students also read a short passage on how Kenyan runners (e.g., marathoners training at altitude in Iten) produce lactic acid during intense exercise, causing temporary muscle fatigue.	Students learned that: (1) the stomach produces HCl (pH 1–2), which activates pepsin and kills pathogens in food such as nyama choma or githeri; (2) the pancreas secretes alkaline bicarbonate to neutralise stomach acid in the duodenum, protecting the intestinal lining; (3) during vigorous exercise, lactic acid accumulates in muscles, lowering blood pH slightly — the body buffers this with bicarbonate ions; (4) alkaline sodium hypochlorite is used as a bleach and disinfectant because OH^--rich environments disrupt microbial cell membranes. Pedagogical note: Use the Kenyan athlete context deliberately — students in Rift Valley schools often have personal connections to athletics, making the lactic-acid example highly motivating.	The anchoring phenomenon is extended: just as lemon juice (acid) breaks down the aluminium oxide on the sufuria, HCl in the stomach breaks down food proteins — controlled acid action is essential for life. Equally, the alkalinity of Lake Magadi (like bleach) makes it hostile to most organisms but useful industrially (Tata Chemicals Magadi extracts soda ash for glass-making). Students update the DQB cluster 'Why do they react with metals/food/living things?' Teachers should pose the question: 'If the stomach is so acidic, why doesn't it digest itself?' — allow 15-second think time, then cold-call 3 students before providing the explanation (mucus lining / bicarbonate barrier).
Lesson 4 **Neutralisation Reactions — How Acids and Bases Cancel Each Other Out**	Students performed a titration-style investigation: they added wood-ash solution (alkali) drop by drop to a measured volume of lemon juice containing universal indicator, swirling after each addition and recording the colour change. They plotted a simple colour-vs-drops graph. A parallel demonstration showed an antacid tablet (magnesium hydroxide or calcium carbonate) dissolving in dilute HCl — students observed fizzing (CO_2 release) and temperature change using a	Neutralisation is the reaction between an acid and a base to form a salt and water: Acid + Base → Salt + Water. The general ionic equation is $H^+ + OH^- \rightarrow H_2O$. When carbonate or bicarbonate bases are used, CO_2 is also produced: e.g., $2HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$. Neutralisation is exothermic (temperature rises). Antacids work by neutralising excess HCl in the stomach. Liming acidic soil raises pH, making nutrients more available to maize and bean crops. Pedagogical note:	The sufuria phenomenon: adding a pinch of bicarbonate of soda to lemon juice before cooking in an aluminium pot neutralises some acid, reducing corrosion — this is a real household practice worth validating scientifically. The Lake Magadi context: industrial acid waste cannot simply be dumped near Magadi because the highly alkaline lake water would neutralise it, producing large quantities of salt — an environmental management consideration. Students update the DQB: 'How do we USE

	thermometer. A third observation: farmers in the Rift Valley add lime (calcium oxide/hydroxide) to acidic red soil — students examined before/after photographs of crop yields from KALRO (Kenya Agricultural and Livestock Research Organisation) trial plots.	Students frequently confuse 'neutralisation always produces pH 7' — clarify that the final pH depends on the relative strengths of the acid and base; a strong acid + weak base gives a slightly acidic salt solution.	acid-base chemistry?' cluster. Cold-call 5 students with: 'Write the word equation for what happens when you take a Maalox tablet for stomach pain — then explain it to your neighbour.'
Lesson 5 **Properties and Preparation of Salts — Beyond Table Salt, from Nairobi Pharmacy to Magadi Flats**	Students examined six salt samples: NaCl (table salt), $CaCO_3$ (chalk/limestone — shown in photographs of Kenyan cement factories in Athi River), Na_2CO_3 (soda ash from Lake Magadi), $CaSO_4$ (plaster of Paris — used in Kenyan hospitals for fractures), NH_4NO_3 (CAN fertiliser used by Kenyan farmers), and $FeSO_4$ (iron(II) sulfate — used in iron-supplement tablets). They tested each for: solubility in water, effect on pH paper (acidic, basic, or neutral salt), and reaction with dilute acid or dilute alkali. Observations were recorded in a structured table. Students also watched a short video clip on the Tata Chemicals Magadi soda-ash extraction process.	A salt is an ionic compound formed when the H^+ of an acid is replaced by a metal ion (or ammonium ion). Salts can be prepared by: (1) neutralisation (acid + base); (2) direct combination of a metal with a non-metal (e.g., $Na + Cl_2 \rightarrow NaCl$); (3) reaction of a metal with an acid (e.g., $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$); (4) reaction of a metal oxide/carbonate with an acid. Salt solutions are not always neutral — they can be acidic (e.g., NH_4Cl) or alkaline (e.g., Na_2CO_3) due to hydrolysis. The Lake Magadi trona ($Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$) is a naturally occurring salt deposit of enormous economic value to Kenya. Pedagogical note: The variety of salts from everyday Kenyan contexts (fertiliser, cement, plaster, supplements) powerfully corrects the misconception that 'salt = NaCl only.'	The anchoring phenomena reconnect: the white residue left on the sufuria after alkaline wood-ash water evaporates is a salt (potassium carbonate, K_2CO_3); the white crust on Lake Magadi's shores is a salt (trona/Na_2CO_3). The corrosion of the sufuria by acid also produces a salt (aluminium salt in solution). Students update the DQB: 'What IS produced when acids and bases react?' is now answered. Teachers cold-call 4 students: 'Name ONE salt used in your home or on your farm, state how it was made (even if industrially), and say whether its solution is acidic, neutral, or alkaline — and why.'
Lesson 6 **Applications of Salts in Real Life — From Lake Magadi to Your Kitchen, Farm and Clinic**	Students rotated through four application stations (10 minutes each): Station A — Food & Preservation: tasting (safely) and testing pH of cured fish from Lake Victoria (salt-preserved), pickled vegetables (acidic brine), and baking-soda chapati dough versus plain dough (observing rise). Station B — Agriculture: comparing maize seedling growth in three soil types — acidic (pH 5), neutral (pH 7), and over-limed (pH 9) — using photographs from a KALRO 8-week trial; examining CAN and DAP fertiliser packets and reading NPK labels. Station C — Medicine: examining an antacid tablet, oral rehydration salts (ORS) packet, and iron-supplement tablet; calculating the moles of active salt in each. Station D — Industry: examining photographs of Athi River cement works, Magadi soda-ash export containers, and Kenyan soap-	Salts are not just a single substance (NaCl) but a vast family of ionic compounds; each salt's specific properties determine its application. Key applications: (1) NaCl — food preservation and flavour; (2) Na_2CO_3 (soda ash) — glass manufacturing, soap production, water softening; (3) $CaCO_3$ — cement and building materials (Athi River Portland Cement); (4) NH_4NO_3 / $(NH_4)_2HPO_4$ (CAN/DAP) — crop fertilisers that replenish soil nitrogen and phosphorus; (5) $Mg(OH)_2$ / $CaCO_3$ — antacids; (6) $NaHCO_3$ — baking agent (releases CO_2 to leaven chapati/mandazi); (7) $FeSO_4$ — iron-deficiency treatment. Pedagogical note: Station C (medicine) is particularly powerful for students from communities where antacids and ORS are common household items — connect the chemistry to lived experience explicitly.	The anchoring phenomena are fully revisited at this stage: wood ash (K_2CO_3, a base) reacts with grease (saponification, producing a soap-like salt) — this is why ash water has historically been used for cleaning in Kenyan rural kitchens. Lake Magadi's soda-ash salts are Kenya's most significant mineral export. Cooking salt (NaCl) is neutral and does not corrode the sufuria. Students update the DQB: the 'Are they safe?' and 'How do we USE them?' clusters are substantially answered. Teachers cold-call 6 students using: 'Pick any one salt from today's stations. Explain its chemical origin (what acid + what base made it), one benefit, and one risk of misuse.'

	making (saponification uses NaOH).		
Lesson 7 **Acid-Base Indicators, Buffers and Soil pH Management — Kenyan Farmer as Chemist**	Students prepared natural indicators from locally available materials: red cabbage juice, turmeric powder, and bougainvillea petal extract (all obtainable at Kenyan markets). They tested each indicator against: dilute HCl, lemon juice, tap water, bicarbonate of soda, wood-ash solution, and dilute NaOH — recording colour changes in a results table. They then compared their natural indicators to universal indicator and discussed sensitivity and range. In the second part, students examined soil pH data from three Kenyan farming regions: acidic pyrethrum soils in Kericho (pH ≈ 5.0), neutral maize soils in Trans Nzoia (pH ≈ 6.8), and slightly alkaline semi-arid soils near Kajiado (pH ≈ 7.8). They calculated how much agricultural lime ($CaCO_3$) a Kericho farmer would need to raise soil pH to the optimal range for pyrethrum (pH 5.5–6.5).	Indicators are substances that change colour depending on pH; they themselves are weak acids or bases whose conjugate pairs have different colours. Natural indicators are culturally and practically significant — Kenyan farmers and food processors have long used colour-change observations (e.g., turmeric staining) without knowing the underlying chemistry. Buffers are solutions that resist changes in pH when small amounts of acid or base are added; blood (pH 7.35–7.45) is buffered by the carbonic acid/bicarbonate system. Soil pH determines nutrient availability: at pH < 5.5, aluminium and manganese become toxic to maize roots; at pH > 7.5, iron and zinc become insoluble. Liming is the most cost-effective intervention for acidic Kenyan highland soils. Pedagogical note: The calculation task (moles of $CaCO_3$ needed per hectare) integrates stoichiometry from earlier units — flag this cross-strand connection explicitly for students preparing for KCSE.	The anchoring phenomenon deepens: the sufuria corrodes fastest in the uji (fermented porridge) because lactic and acetic acid lower pH to ≈ 3.5 — even below lemon juice on some days, depending on fermentation time. The Lake Magadi water (pH ≈ 11) acts as a natural buffer due to the carbonate/bicarbonate system present in the lake — this is why its pH remains remarkably stable despite rain and evaporation cycles. Students update the DQB: 'How do we MEASURE and CONTROL pH in real systems?' is now answered. Teachers pose the synthesis question: 'A Kericho tea farmer notices her pyrethrum crop failing. She tests the soil and gets pH 4.8. Walk me through, step by step, the chemistry she needs to fix this.' Cold-call 3 students for sequential steps.
Lesson 8 **Final Explanation: Environmental Impact, Community Project Showcase & Assessment — Closing the Loop on Lake Magadi and the Sufuria**	Students presented community-action mini-projects (prepared in Lessons 6–7 as homework/group work) addressing one of three real Kenyan scenarios: (A) advising a Kisumu fishing community on how over-application of CAN fertiliser causes eutrophication in Lake Victoria; (B) designing a safe household-waste disposal guide for a Nairobi estate covering acids (battery acid, drain cleaners), bases (bleach, oven cleaner), and neutral salts (table salt, ORS); or (C) proposing a school garden soil-management plan for a Rift Valley school using liming and composting to maintain optimal pH. Each group displayed a poster and gave a 3-minute presentation. The class also completed a final individual written assessment (structured questions + one extended-response question returning to the original anchoring phenomena).	Students synthesised the full unit: (1) excess fertiliser salts (CAN, DAP) dissolve in runoff water, enter Lake Victoria, cause eutrophication — algal blooms consume dissolved oxygen, killing fish that Kisumu fishing families depend on; (2) improper disposal of acidic or alkaline household chemicals can corrode pipes, harm soil organisms, and contaminate groundwater; (3) sustainable agriculture requires monitoring and adjusting soil pH using liming (for acidic soils) or sulfur/organic matter (for alkaline soils); (4) the sufuria corrodes because acids donate H^+ to aluminium, forming Al^{3+} ions and H_2 gas — understanding this allows households to make informed choices about cookware and food storage; (5) Lake Magadi's economic value (soda-ash export) must be balanced against ecological protection of the flamingo habitat. Pedagogical note: The	The driving question is fully answered and closed: substances from the Kenyan environment react differently because of their H^+/OH^- ion concentrations and the specific salts they contain; understanding acid-base chemistry allows communities to protect cookware, manage soil, treat illness, preserve food, run industries (Magadi soda ash), and protect ecosystems (Lake Victoria eutrophication prevention). The DQB is reviewed as a class — every original question from Lesson 1 is matched to the lesson(s) that answered it, demonstrating the full arc of learning. Teachers should display both the Lesson 1 and Lesson 8 versions of the DQB side by side and cold-call 5 students with: 'Point to one question on the Lesson 1 board that YOU can now fully answer — and give that answer in two sentences using chemistry vocabulary.'

		final assessment should be marked using a concept-map rubric that rewards connections between lessons, not just recall of isolated facts. Compare students' final DQB entries to their Lesson 1 sticky notes — celebrate measurable conceptual growth explicitly in class.	